SIMATS ENGINEERING

Department of Computer Science and Engineering

CSA51 – Cryptography and Network Security

ASSIGNMENT 3 – CO3

Examine Cryptographic Hash Algorithms, Digital Signatures, Kerberos and X.509 Authentication Frameworks

CO3: Examine cryptographic hash algorithms and digital signature schemes, and assess Kerberos and X.509 authentication frameworks for ensuring integrity, authentication and non-repudiation.

Student Name	Register Number	Department / Section	Academic Year / Semester	Faculty Name	Nature	Duration	Total Marks	Date of Submission	Date of Evaluation
				Dr. R. Dharani	Pair / Individual	As assigned	50		

MARK SHEET (Rubric-based Evaluation)

Task	Task Title	Max Marks	Marks Awarded	Remarks / Evidence Expected
1	Cryptographic Hash Properties, Avalanche Effect, Collision Resistance and Birthday Bound	6		Hash properties and attack-scenario explanations; Toy hash versus SHA-256 comparison and fixed-length digest reasoning; Avalanche experiment design, bit-change measurement and analysis; Truncated-hash collision search, birthday-bound derivation and evidence
2	Hash Functions, MAC and HMAC – Construction and Security Analysis	7		Toy Merkle–Damgård hash and padding/chaining implementation; Plain-hash integrity limitation and attacker-controlled digest demonstration; Naive H(key ‖ message) and length-extension attack; HMAC inner/outer construction, tamper tests and security explanation; Hash vs. MAC vs. HMAC comparison and joint design decision
3	Digital Signatures – RSA, DSA/ECDSA, Verification and Attack Analysis	7		RSA hash-then-sign implementation and mathematical verification; DSA/ECDSA derivation and toy implementation; Valid/invalid signature and wrong-public-key test cases; Nonce-reuse private-key recovery and textbook RSA forgery demonstration; Non-repudiation conditions, key compromise and trusted timestamp/third party
4	X.509 Certificates, CA, Chain Validation and Revocation Mechanisms	7		X.509 v3 fields, extensions and three-level certificate hierarchy; Chain-validation algorithm/pseudocode and trust-anchor explanation; At least six certificate failure-case tests and rejection evidence; CRL vs. OCSP analysis and revocation/fail-open policy; CA compromise/mis-issuance impact and mitigation decision
5	Kerberos Authentication, Ticket-Granting System, Replay Attacks and Mutual Authentication	6		AS, TGS and client-service message-sequence diagram with key labels; Ticket, session key, authenticator, timestamp and mutual-authentication explanation; Replay, expired-ticket, clock-skew and KDC-unavailable experiments; Stolen-ticket/offline password-guessing risks and cross-understanding
6	Advanced Digital Signature Schemes and Kerberos vs. X.509 Comparison	7		Collision-to-signature-forgery investigation and hash-policy recommendation; RSA, DSA/DSS, ECDSA, ElGamal and Schnorr comparison; Schnorr commitment–challenge–response and verification relation; ElGamal key generation, signing, verification and altered-case tests; Kerberos vs. X.509, key-relationship scaling and use-case recommendations
7	Integrated Design, Implementation, Security Validation and Evaluation of Authentication and Integrity Scheme	10		Independent Student A and B proposals retained; Comparison record, threat model and security goals; Final protocol/message-flow, key/certificate management and pseudocode; Security argument, attack-effort estimates and residual risks; Implementation, baseline comparison, attack/correctness tests and measured costs; Cross-defence and final engineering evaluation/trade-offs
	TOTAL	**50**		**Marks to be awarded by the faculty.**

OVERALL REMARKS

Faculty feedback: _____

_____

Student A Signature	Student B Signature	Faculty Signature	Programme Director

MANDATORY SUBMISSION EVIDENCE CHECK

✓ = Present P = Partly Present ✗ = Missing. Mark the status after checking the submitted report, code and experimental evidence.

Question	Evidence Expected	Status	Note / Verification
1	Toy hash and SHA-256 comparison; avalanche bit-flip inputs and changed-bit counts; at least three message sizes; truncated-hash collision runs for k = 16, 20, 24, 28, 32; tables/graphs and birthday-bound explanation.	☐	Evidence must be traceable to the report, source code or experimental output.
2	Toy Merkle–Damgård hash; plain hash, H(key || message) and HMAC code; modified-file/digest, appended-data and length-extension traces; tamper accept/reject table; mechanism-selection justification.	☐	Evidence must be traceable to the report, source code or experimental output.
3	RSA and DSA/ECDSA signature code; derivations; valid and altered-message/signature tests; wrong-key tests; nonce-reuse recovery trace; textbook RSA forgery evidence; non-repudiation discussion.	☐	Evidence must be traceable to the report, source code or experimental output.
4	Root/intermediate/end-entity certificate chain; validation pseudocode; valid and invalid chain outputs; at least six failure cases; CRL/OCSP comparison and revocation-policy analysis.	☐	Evidence must be traceable to the report, source code or experimental output.
5	Kerberos AS–TGS–service message-sequence diagram; key-labelled messages; simulation output; replay, expired-ticket, clock-skew and KDC-outage results; stolen-ticket/password-guessing analysis.	☐	Evidence must be traceable to the report, source code or experimental output.
6	Collision-to-forgery toy example; signature-scheme comparison table; Schnorr derivation; ElGamal worked example and tamper tests; Kerberos/X.509 comparison and key-relationship analysis.	☐	Evidence must be traceable to the report, source code or experimental output.
7	Both independent proposals; comparison record; threat model; final architecture and protocol diagram; signing/verification pseudocode; security argument; implementation logs; baseline comparison; validation results and residual-risk evaluation.	☐	Evidence must be traceable to the report, source code or experimental output.

IMPLEMENTATION EVIDENCE CHAIN(Task 5-7)

Evidence Item	Status	Observation
Source code / algorithms	☐	Readable student-developed code, language/dependencies and execution instructions are provided.
Test inputs and parameters	☐	Messages, bit-flip inputs, toy keys, nonces, certificates, tickets and test cases are recorded.
Mathematical derivations	☐	Avalanche expectation, birthday bound, signature equations, nonce recovery and protocol calculations are shown.
Predictions and expected behaviour	☐	Expected digest changes, collisions, signature outcomes, certificate decisions and Kerberos responses are stated.
Actual output	☐	Program outputs, logs or screenshots show reproducible results.
Attack evidence	☐	Toy-only length extension, nonce reuse, RSA forgery, certificate failure, replay and collision-to-forgery traces are included.
Performance measurements	☐	Hash counts, collision-search effort, verification operations, execution time and metadata costs are tabulated.
Boundary / failure tests	☐	Altered messages/signatures, wrong keys, expired/revoked certificates, replay and clock-skew cases are tested.
Interpretation	☐	Results are linked to integrity, authentication, non-repudiation and trust assumptions.
Consistency / reproducibility	☐	Code, raw data, tables, diagrams and report conclusions agree and can be rerun.

SUBMISSION CHECKLIST (not separately marked)

Requirement	Yes	Partly	No
All questions Q1–Q7 and all specified individual/shared responsibilities are addressed.	☐	☐	☐
Student A and Student B individual contributions are clearly identified.	☐	☐	☐
Both students can explain and defend the complete work, not only their assigned part.	☐	☐	☐
Hash properties, avalanche measurements and birthday-bound analysis are included.	☐	☐	☐
Hash-only, naive MAC and HMAC constructions are implemented and compared.	☐	☐	☐
RSA and DSA/ECDSA signatures, nonce risks and forgery tests are evidenced.	☐	☐	☐
X.509 chain validation and certificate revocation/failure cases are demonstrated.	☐	☐	☐
Kerberos message flow, ticket handling, replay protection and failure cases are tested.	☐	☐	☐
RSA, DSA/DSS, ECDSA, ElGamal and Schnorr comparison is included.	☐	☐	☐
Q7 independent proposals, comparison record and final joint design are retained.	☐	☐	☐
Raw outputs, calculations, test cases, tables/graphs and protocol diagrams are included.	☐	☐	☐
Source code and execution instructions are reproducible; report claims match outputs.	☐	☐	☐
Only toy parameters and synthetic test data are used for attack demonstrations.	☐	☐	☐
No unfilled placeholders remain; report numbering and references are consistent.	☐	☐	☐

Safety note: All attack demonstrations must use only student-created toy parameters and synthetic test data. Do not use real credentials, private keys or production certificates.

Assessment note: Faculty verification/viva may assess individual contribution, cross-understanding, mathematical derivations, source code, attack traces, experimental evidence and the Q7 final design.